

(ii) $A = \{(1,4), (3,5), (5,3)\}$
 $B = \{ \}$

20/07/23

(TOC) theory of Computation.

Formal Language & Automata Theory

→ Language, Grammar, Machine.

LANGUAGE :-

- Σ = Finite set of input-symbols is known as Alphabet.
 set of alphabet
~~(finite set of)~~
 $\Sigma = \{a, b, c\}$
- Strings : Collection of Symbols will make a String.
 'ε' : 0 length string, which is the smallest string length string.
 → whether the string creates a meaning or not, it's not considerable.
 → Sequence of character matters.
 $abba \neq aabb$
 $aab = a^2b$
- if $\Sigma = \{a, b\}$ then set of strings possible,
 $\{ \epsilon, a, b, aa, bb, aba, abba, baba, \dots, \infty \};$

Language

Language :- Collection of string makes a language over some rules.

L1: All the strings, starting with 'a' where $\epsilon \in \Sigma = \{a, b\};$
 $= \{a, aa, ab, \dots\}$
 $= \{a^n, ab, abb, \dots, \infty\}$

- Language is a subset of total no. of string possible in Σ .

L2: All strings ending with 'ab', where $\Sigma = \{a, b\}$.

$$= \{a^n, b^n, a^n b^n ab\}$$

$$= \{ab, aab, abab, bab, \dots\}$$

L3: All strings of length exactly 3, where $\Sigma = \{a, b\}$

$$= \{aaa, bbb, aba, abb, bba, \dots\} \text{ finite}$$

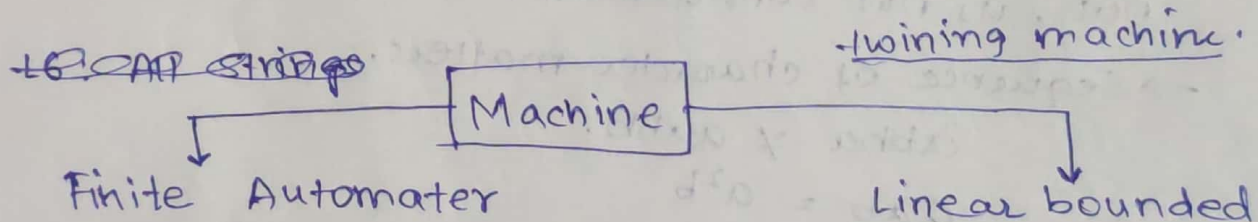
* L4: All strings, where no. of 'a' is divisible by 2
 $\Sigma = \{a, b\}$.

$$= \{\epsilon, a^2, a^{2n}b, \underline{aba}, \dots\} \text{ infinite}$$

* L5: All strings, ~~binary~~ ^{decimal} converted value is div by 5.

$$\Sigma = \{0, 1\}$$

$$= \{\epsilon, 101, 0, 000, 0^n, 1111, 1010, \dots\}$$



Σ^0 : What are strings possible using Σ , 25/07/23
~~using~~ of length

$$\Sigma^0 = \{\epsilon\}$$

$$\Sigma^1 = \{a, b\}$$

$$\Sigma^2 = \{aa, bb, ab, ba\}$$

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots = \{\epsilon, a, b, aa, ab, ba, bb, \dots\}$$

Σ^* : By using Σ , various string possible.

↳ Kleene Closure.

$a^* = a^0 \cup a^1 \cup a^2 \dots \infty$ [always infinite].

$$\Sigma^* = (a+b)^*$$

a^+ : positive closure : All possible strings excluding a^0 . (ϵ not included).

$$\Sigma^+ = \Sigma^* - \epsilon$$

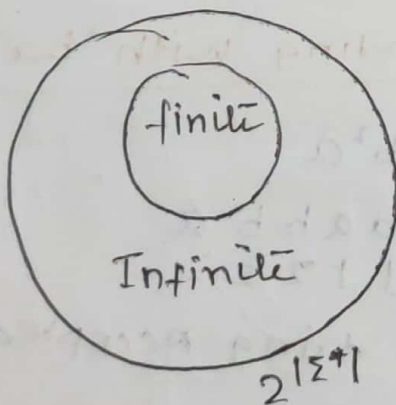
$\Sigma = \{a, b, c\}$.

$\Sigma^* = \{\epsilon, a, b, c, aa, bb, ab, bc, ccc, \dots\}$

$L \subseteq \Sigma^*$ any lang. is a subset of Σ^*

By using Σ the total no. of lang possible is $2^{|\Sigma^*|}$
 $2^{|\Sigma^*|}$ Power set conc.

VEN DIAGRAM :-



L1 : $\forall$ strings starting with 'a', $\Sigma = \{a, b\}$.
 $= \{a, a^2, ab, aba, \dots, \infty\}$

Machine : Finite Automata.

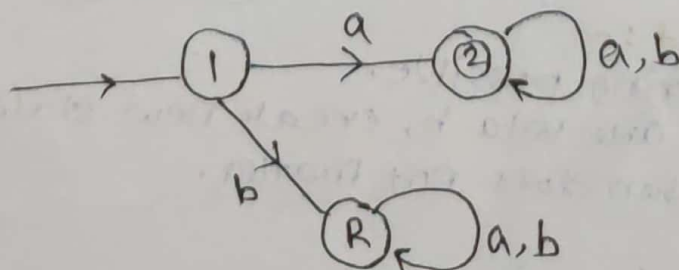
$Q = 1, 2, R$

$\Sigma = a, b$

$q_s = 1$

$F = \{2\}$

$\delta(1, a) = 2$



TRANSITION DIAGRAM.

FA
5 tuples

Q set of states

$Q, \Sigma, \delta, q_s, F$

$Q \rightarrow$ finite set of states.

$\Sigma \rightarrow$ set of alphabet

$\delta \rightarrow$ transition fn.

$q_s \rightarrow$ starting state

$F \rightarrow$ set of final states

Starting state !

- * Only one starting state
- * Starting state will have transition but no input.

$$q_s = 1 \subseteq Q$$

- * Q finite set of states.

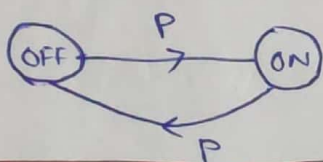
q_s can also be F .

Finite state can be more than 1.

abbab
1 2 2 2 2 2

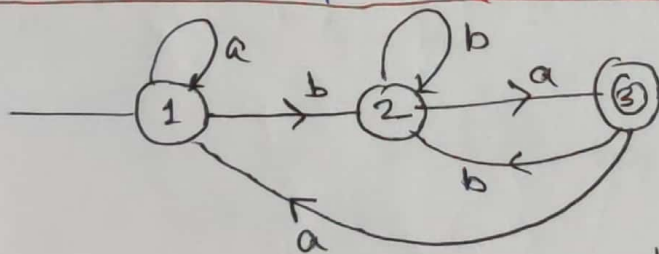
Hence accepted.

WORKING OF SEILING FAN



$\Sigma = \{P\}$.

All strings Ending with 'ba' $\Sigma = \{ab\}$



a^3b^2a

aaabba

1 1 1 2 2

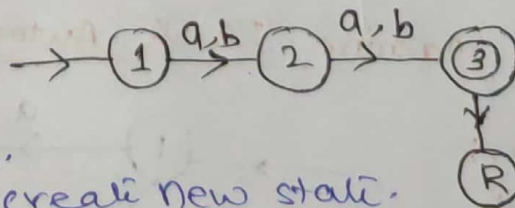
hence string Accepted.

a^3ba^2
1 2 3 1

hence Rejected.

$L1$: All strings length exactly 2 $\Sigma = \{a, b\}$.

$= \{a^2, ab, ba, b^2\}$.



Skeleton Method:-

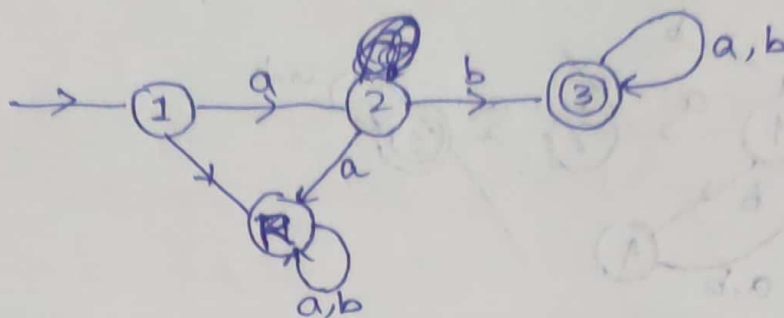
→ Smallest string possible.

→ jblge result are wla h, create new state.

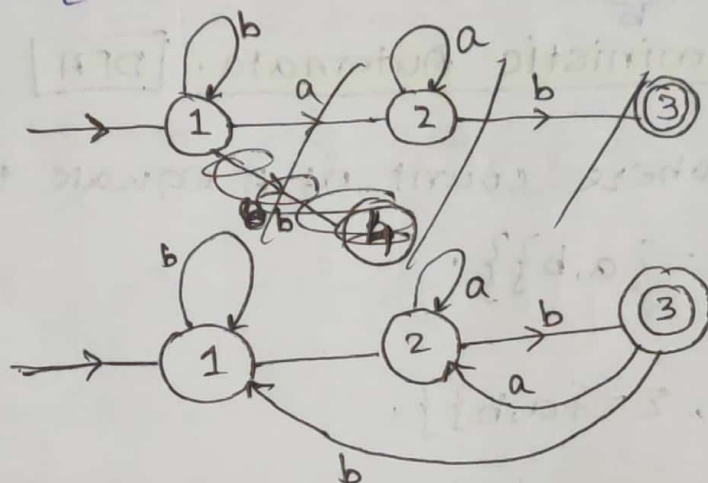
→ self loop when does not matter.

$$= \emptyset (a+b)(a+b)$$

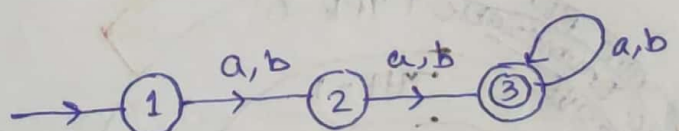
L2: All strings starting with 'ab', $\Sigma = \{a, b\}$.
 $= \{ ab, abba, abbb, ab\Sigma^* \}$.



L3: All strings ending with 'ab', $\Sigma = \{a, b\}$.
 $= \{ ab, aab, bab, a^2ab, \dots, \Sigma^*ab \}$



L4: All strings with length atleast 2, $\Sigma = \{a, b\}$.
 $= \{ aa, ab, ba, bb, aaa, \dots \}$, $\Sigma\Sigma\Sigma^*$ or $\Sigma^2\Sigma^*$

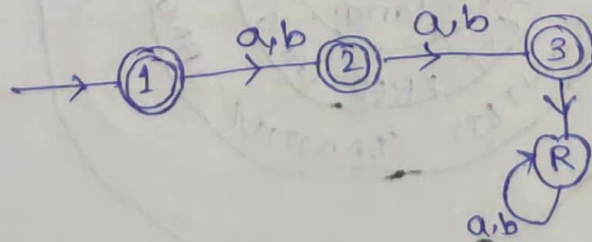


$$= (a+b)(a+b)^+$$

* count no self loop.

L5: All strings atmost 2, $\Sigma = \{a, b\}$

$= \{ \epsilon, a, b, aa, ab, ba, bb \}$



* If lang is finite then it will contain no self loop and requires reject state.

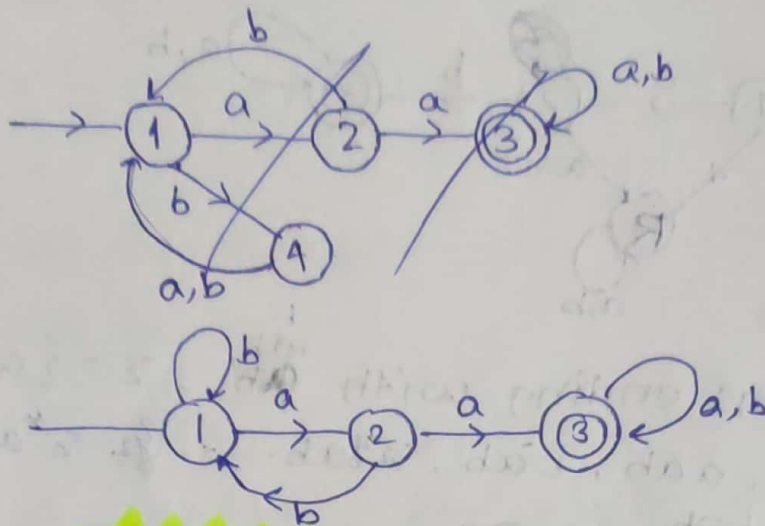
* length atleast n total states = n+1

* length atmost (n+2)

SUBSTRING:

abba $\rightarrow \epsilon, a, ab, abb, abba.$

L6: All string having aa as a substring, $\Sigma = \{a, b\}$
 $= \{aa, aaa, aaba, \dots, \Sigma^*aa\Sigma^*\}$

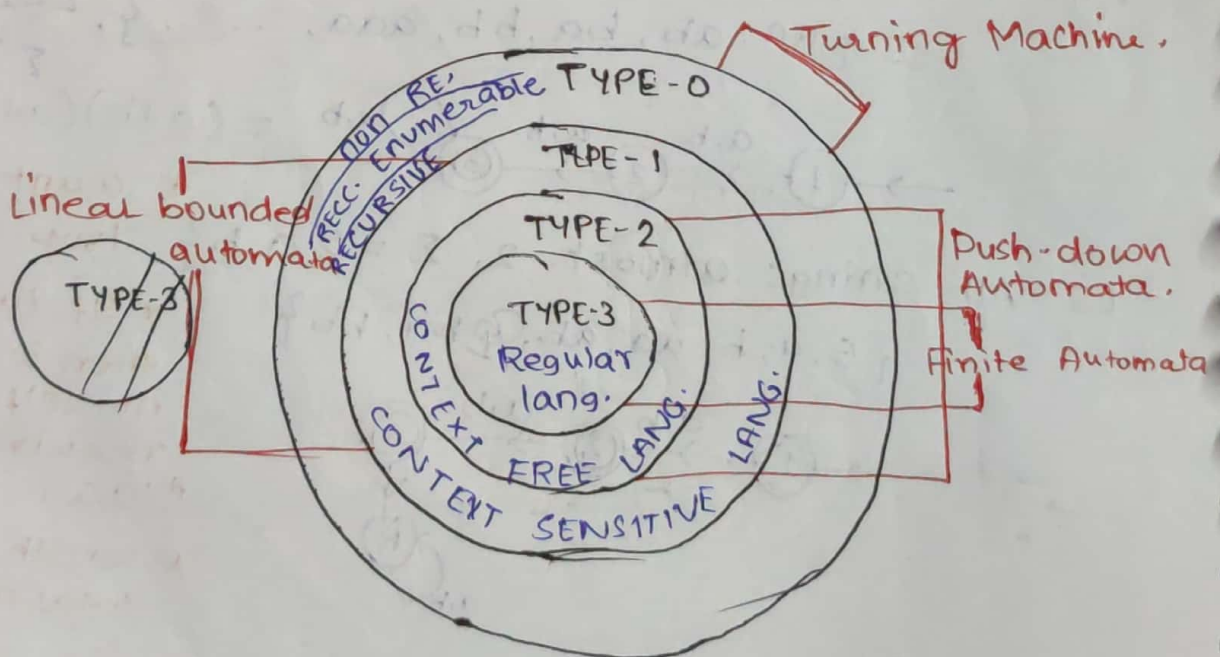


Deterministic Automata. DFA

L7: All strings where count of A equals to count of B.
 $\{n_a = n_b ; \Sigma = \{a, b\}\}$

L8: $\{a^n b^n \mid n \geq 1, \Sigma = \{a, b\}\}$

Chomsky's Classifications:-



L is Regular lang. if and only there exist a finite automata for that.

Finite Automata :-

* finite language. i.e. lang is Regular.

Note: Type 2, T1 and T0 all are infinite languages.

Construct a DFA for the lang. below?

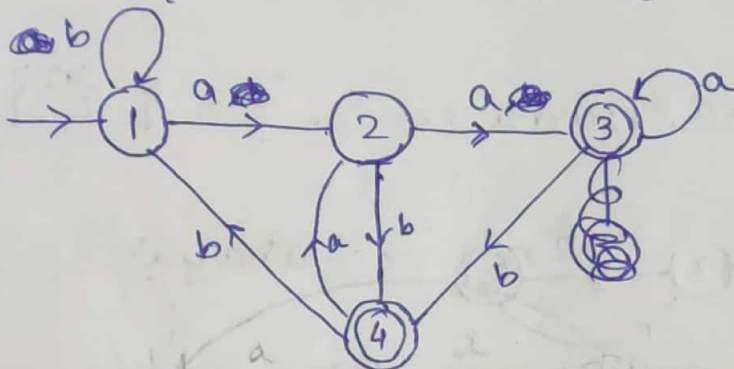
ASSIGNMENT 1.

L9: All strings where 2nd last character will be 'a', $\Sigma = \{a, b\}$

2. L10: 3rd last char will be 'b'.

$$L9 = \{\Sigma^*ab\} \text{ or } \{\Sigma^*a\Sigma\}$$

$$= \{aa, ab, bab, \dots\}$$



if from right side
nth pos is even,
then total state

$$TS = 2^n$$

from left nth pos.

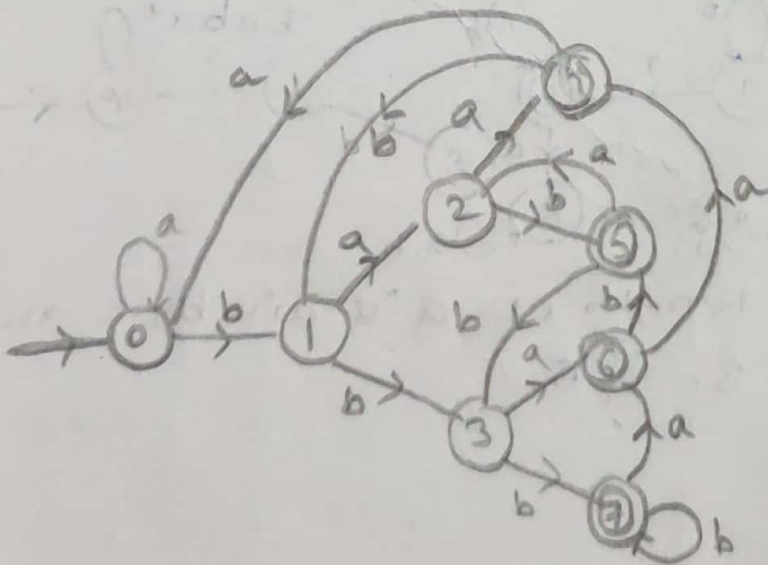
$$TS = n + 2$$

divi by n $TS = n$
 $\forall n = 0 \text{ or } \forall n \neq 0$

L10: $\{ bbbb, bab, abab, \Sigma^*b\Sigma^2 \}$
 $(a+b)^*b(a+b)(a+b)$

No. of states $= 2^n = 2^3 = 8$ $n=3$

No. of final states $= 2^{n-1} = 4$

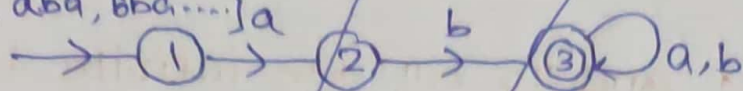


Transition-Table

	a	b → odd
q ₀	q ₀	q ₁
q ₁	q ₂	q ₃
q ₂	q ₄	q ₅
q ₃	q ₆	q ₇
q ₄	q ₀	q ₁
q ₅	q ₂	q ₃
q ₆	q ₄	q ₅
q ₇	q ₆	q ₇

○ final states.

L11: $(a+b)b(a+b)^*$ ~~$= (a+b)^*b(a+b)$~~
 $= \{aa, aba, bba, \dots\}a$

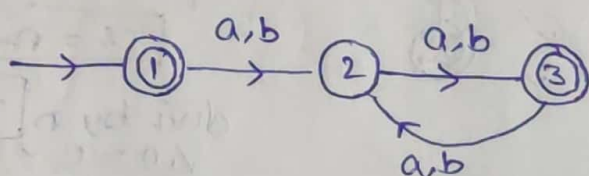


L12: $(a+b)^*b(a+b)$

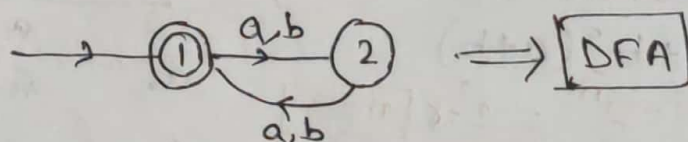
01/08/23

L11: All strings where length is divisible by 2.

$= \{\epsilon, aa, aaaa, \dots\} = \{a^{2n} \mid n=0,1,\dots\}$

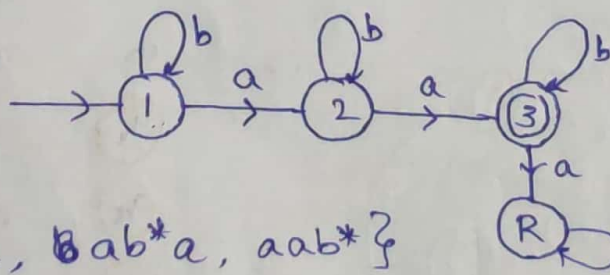


ababab



DFA

L12: All String where no. of a is exactly 2.

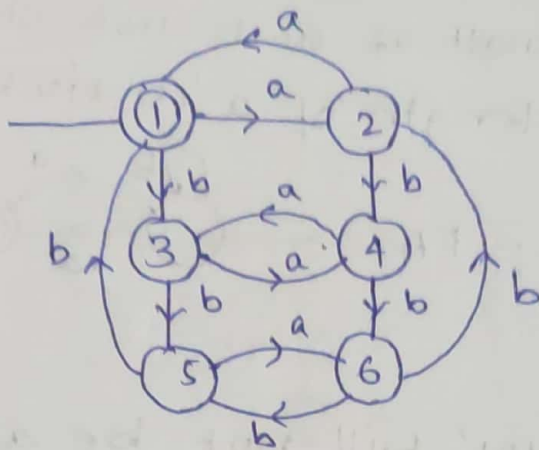


baba

$= \{b^*aa, bab^*a, aab^*\}$

* L13: All strings where length of a's div by 2 and length of b's div by 3. $= \{\epsilon\}$





L14: All strings where length of a is atleast 3 and length of b is atmost 4. $\Sigma = \{a, b\}$

$$\begin{aligned} \text{TS for 'a'} &= 4(n+1) \\ \text{TS for 'b'} &= 6(n+2) \end{aligned}$$

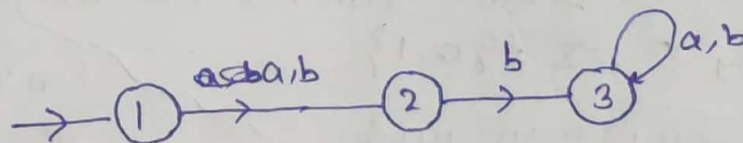
atleast n (or) at-most m then $TS = \max(n, m)$

COMPLIMENT OF A LANGUAGE :- (L^c)

Construct a DFA for the language,

L1: All strings where no b is at 2nd pos. $\Sigma = \{a, b\}$

Create FA of $L1^c$, All strings 'b' at 2nd pos.

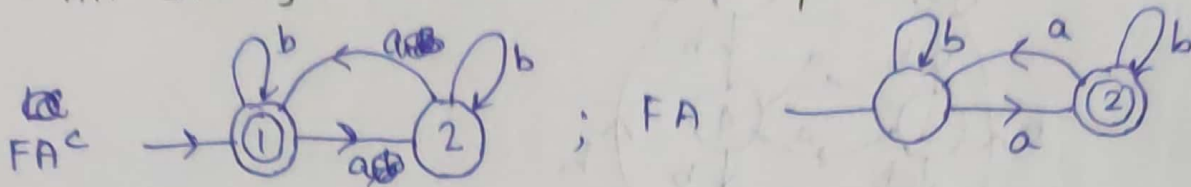


L	L^c
FA	FA
final states	non-final states

~~122~~

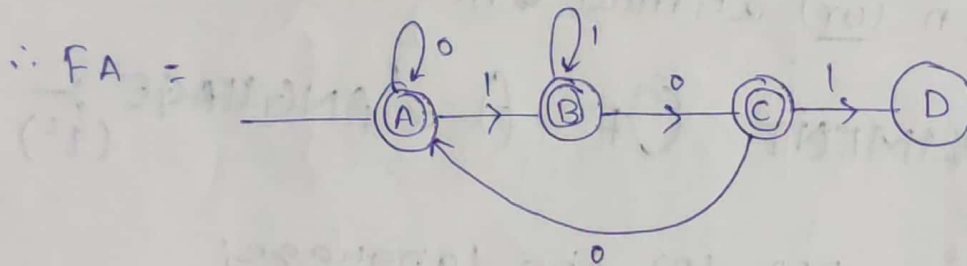
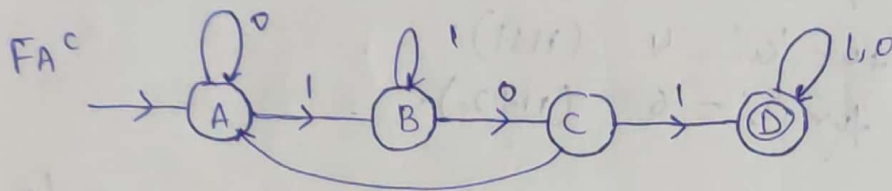
L2: All strings where length of a is not div by 2.

L2^c: All strings where length of a is div by 2.



ASS 3:

L3: All strings where '101' will not be a substring
 $\Sigma = \{0, 1\}$.

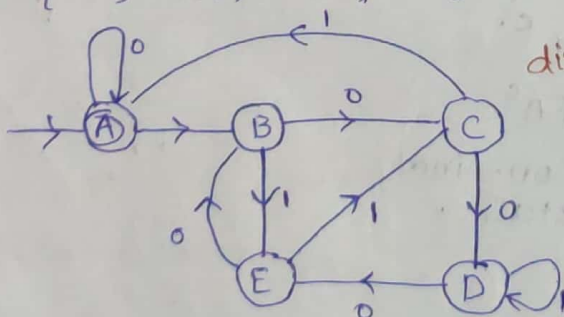


* If 'no something' is given in language 02/08/23
 then it is complementary language.

L1: $\forall$ string whose decimal converted value will be
 divisible by 5, $\Sigma = \{0, 1\}$

$L = \{ \epsilon, 0101, 1010, 1111, \dots \}$

if $n \% 5 = 0$
 no. of states = n

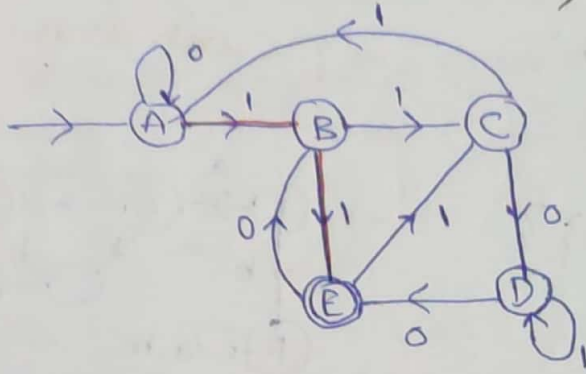


divide by 5

Transition table.

	0	1	→ Σ
q_0	q_5	q_1	
q_1	q_2	q_3	
q_2	q_4		
$\vdots$	$\vdots$		
q_n		q_n	

L : A string whose decimal converted value be divisible by 5 given remainder 3, $\Sigma = \{0, 1\}$



draw same DFA as div by 5 just don't mention the final state.

As remainder is 3 i.e. 11 check & mention final state then it's (E).

$$L : \left\{ \begin{array}{l} 7n+2 \\ 1 \end{array} \middle| \begin{array}{l} \Sigma = \{1\}, \\ n \geq 0 \end{array} \right\}$$

$$7 \times 0 + 2 = \{1^2, 1^9, \dots\}$$

no. of '1' is $\% 7 = 2$
states will be 7.

Regular Grammar

Generates

Regular Language

Accept/Rejected by

Finite Automata

Regular Express

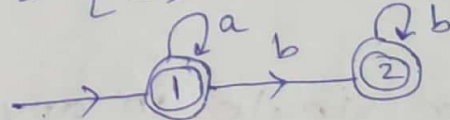
$$L : (a+b)^* \mid \Sigma = \{a, b\}$$

$$= \{\epsilon, a^1, a, b, bb, abb, \dots\}$$



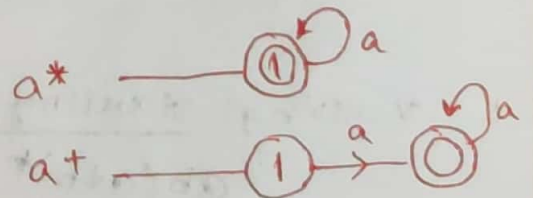
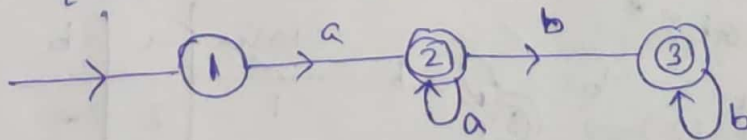
$$L : \{a^*b^* \mid \Sigma = \{a, b\}\}$$

$$= \{\epsilon, a^2, b^2, a^3b^3, \dots, \infty\}$$



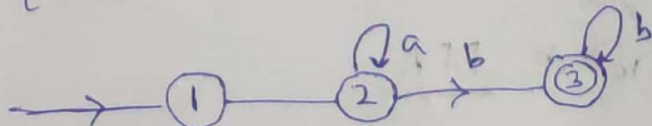
$$L = \{a^+b^* \mid \Sigma = \{a, b\}\}$$

$$= \{a, ab, aa, aabb, \dots, \infty\}$$



$$L = \{a^+b^+ \mid \Sigma = \{a, b\}\}$$

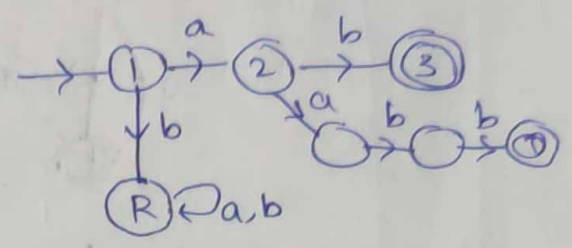
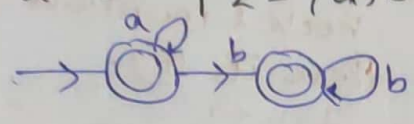
$$= \{ab, aabb, \dots, \infty\}$$



$L = \{ a^n b^n \mid n \geq 1, \Sigma = \{a, b\} \}$

~~a^*b^*~~ = $\{ ab, a^2b^2, \dots \infty \}$ can't draw DFA.

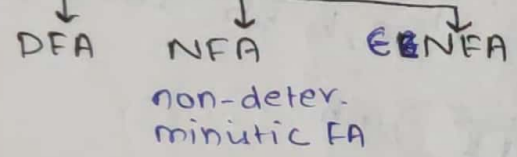
$L = \{ a^*b^* \mid \Sigma = \{a, b\} \}$



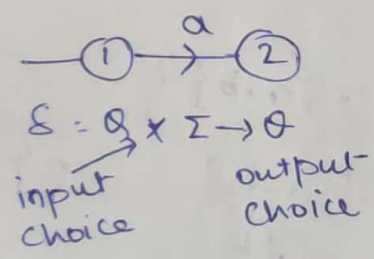
PDA = FA + 1 stack $\rightarrow$ push down Automata.

Finite Automata $[Q, \Sigma, \delta, q_s, F]$

Without output



With output
Moore Machine
Mealy Machine

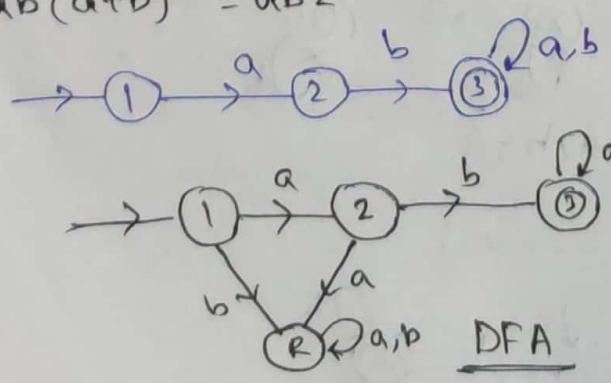


* Conceptually no. of states will be less in NFA.

NFA : For single in any state there are multiple choices.
 $\delta : Q \times \Sigma \rightarrow 2^Q$

L = string starting with 'ab', $\Sigma = \{a, b\}$ Draw NFA.
 $ab(a+b)^* = ab\Sigma^*$

states	a	b
{1}	{2}	R
{2}	R	{3}
{3}	{3}	{3}
{R}	R	R



states	a	b
1	2	-
2	-	3
3	3	3

L & string ending with 'ab' $\Sigma = \{a, b\}$.

$$(a+b)^* ab = \Sigma^* ab$$

* Power of any type FA is same, it is convertible.

$\delta \rightarrow$ transition diagram.

$\rightarrow$ table.

$\rightarrow$ function.

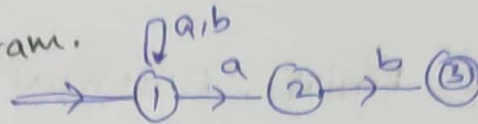


Table NFA

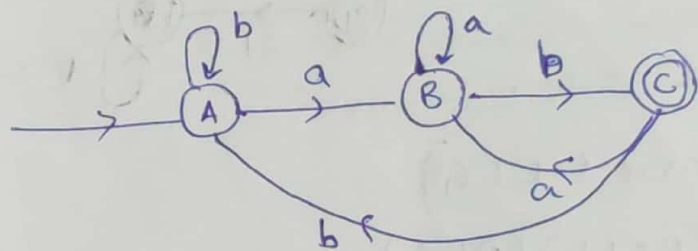
NFA

status	a	b
1	1, 2	1
2	-	3
3	-	-

$$\begin{aligned} \delta(1, a) &= 1, 2 \\ \delta(1, b) &= 1 \\ \delta(2, b) &= 3 \end{aligned}$$

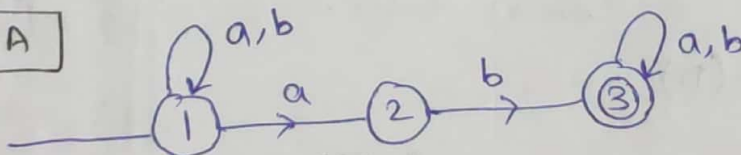
NFA $\rightarrow$ DFA

status	a	b
$\{1\}_A$	$\{1, 2\}_B$	$\{1\}_A$
$\{1, 2\}_B$	$\{1, 2\}_B$	$\{1, 3\}_C$
$\{1, 3\}_C$	$\{1, 2\}_B$	$\{1\}_A$



L & string which with 'ab', $\Sigma = \{a, b\}$, $(a+b)^* ab(a+b) = \Sigma^* ab \Sigma^*$.

NFA



09/08/23

ASSIGNMENT 4- FA which you can't draw. Minimisation method :-

states

(ABCR)

(ABR) (C)

(AR) (B) (C)

(A) (R) (B) (C) R & A are in same set.

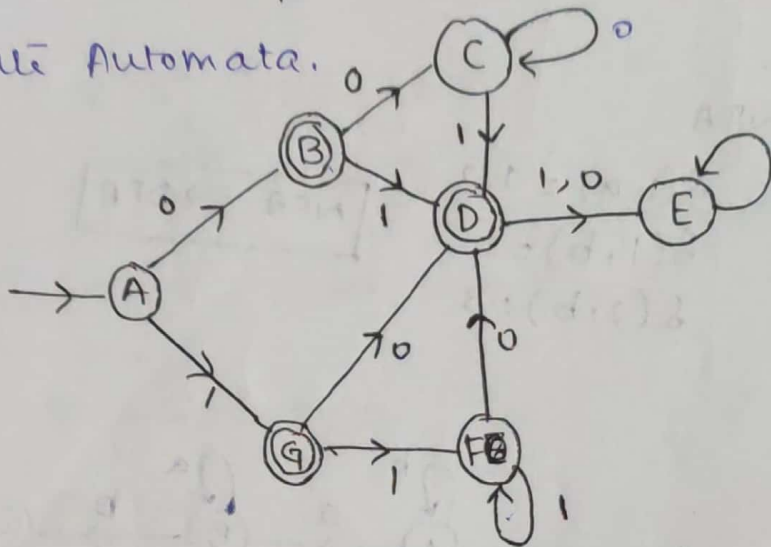
Hence minimisation not possible

- NFA → DFA
- Construct DFA
- Minimisation of FA.

→ FA to regular expression.

Minimisation of Finite Automata.

Finite Automata.



states	0	1
A	B	G
*B	C	D
C	C	D
*D	E	E
E	E	E
F	D	F
*G	D	F

$(ABCDEFGG)$
 $\rightarrow (ACEF)(BDG)$
 $\rightarrow (AE)(CF)(BDG)$
 $\rightarrow (AE)(CF)(BG)(D)$
 $\rightarrow \downarrow$
 $P \quad q \quad r \quad s$

final
status = G, d as
BG & D are
final status.

states.	0	1
*AP	R	R
*AR	B	D/P
B	A	D
D	P	P
P	P	P
Q	D	Q
R	D	F/Q

$(A)(E)(C)(F)(B)(G)(D)$

Regular Expressions

- 1) $\Sigma = \{a\}$, 'a' is a RE
- 2) $\Sigma = \{a, b\}$, a.b is RE
• concatenation.
- 3) $\Sigma = \{a, b\}$, $(a+b)$ is RE
- 4) $\Sigma = \{a, b\}$, ϵ is RE
- 5) Σ^* kleene closure.
 a^* is RE in $\Sigma = \{a\}$.
- 6) Σ^+ a^+ is RE.
 aa^*

~~← NFA~~

$$(a^*b^*)^* = (a+b)^*$$

$(a+b)(a+b)^+$ $\Sigma = \{a, b\}$
length atleast 2.

$(a+b)^+$
 $(\epsilon + (a+b) + (a+b)(a+b) + \dots)$
 $(\epsilon + a + b + ab + ba + bb + \dots)$
 use 1st bracket.

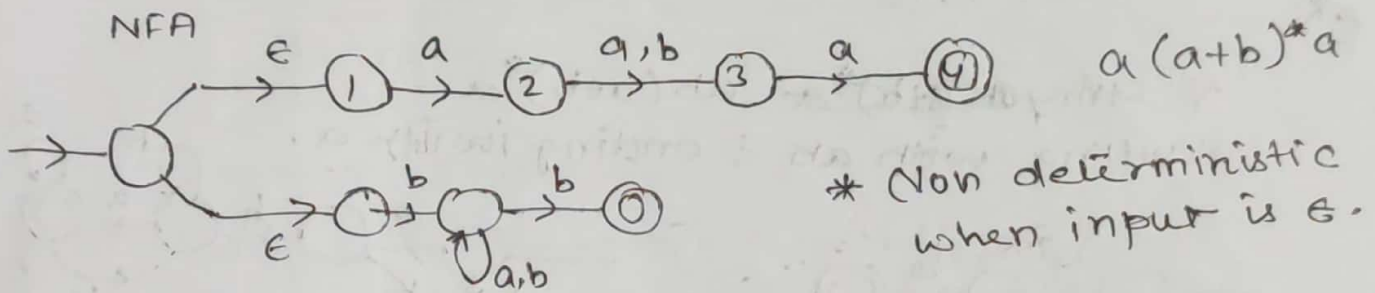
* length div by 2.

$$\{ \epsilon, (a+b)(a+b), (a+b)^4, \dots \}$$

L : starting and ending with same character,
 $\Sigma = \{a, b\}$

$$= \{ a, b, a\Sigma^*a, \cancel{(a+b)\Sigma^*} b\Sigma^*b \}$$

$$((a(a+b)^*a) + (b(a+b)^*b))$$



(ABCD) (EFGHIJ) (R)

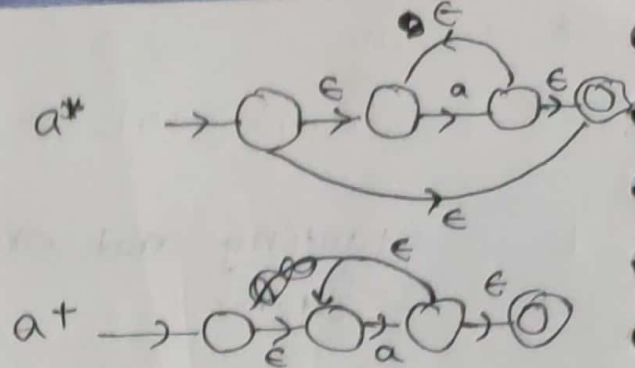
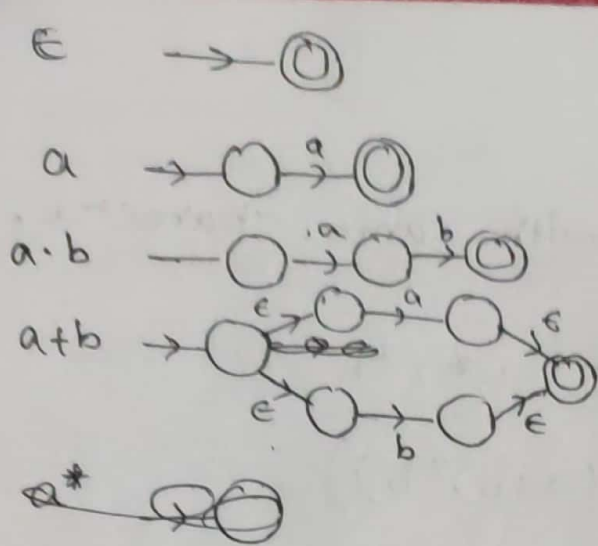
(A) (BDC) (EI) (FGH) (J) (R)

(A)(BCD)(EI)(FGH)(J)(R)

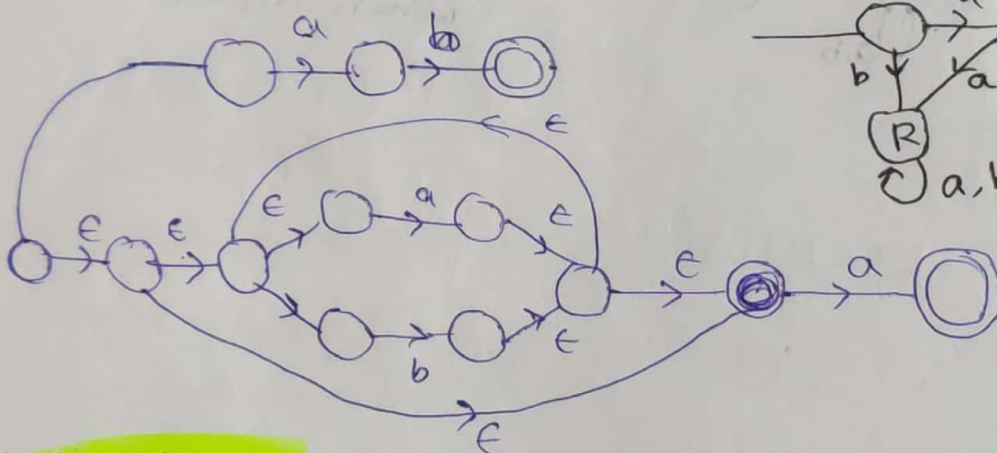
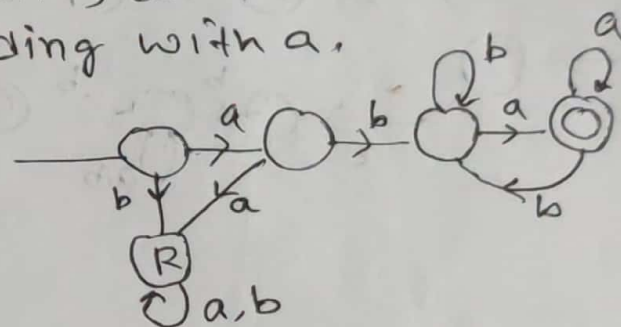
(A) (B) (CD) (EI) (FG) (H) (J) (R)

1, 6, 7,

X 2 3 4 5 6 7



L $\forall$ string $ab(a+b)^*a$ starting with ab & ending with a .

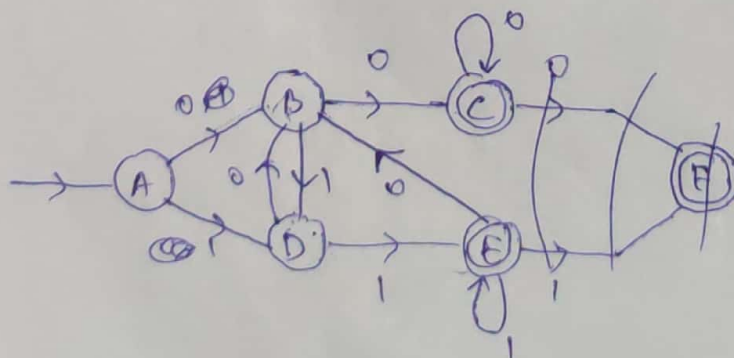


ASSINGMENTS — minimisation. any 1 from grp.

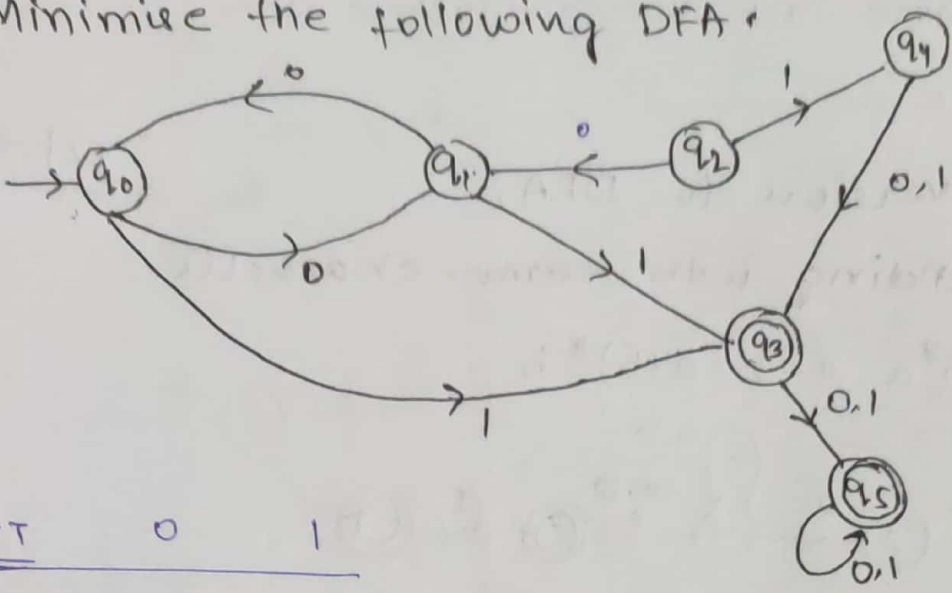
00, 11 string ending.

011

01101100



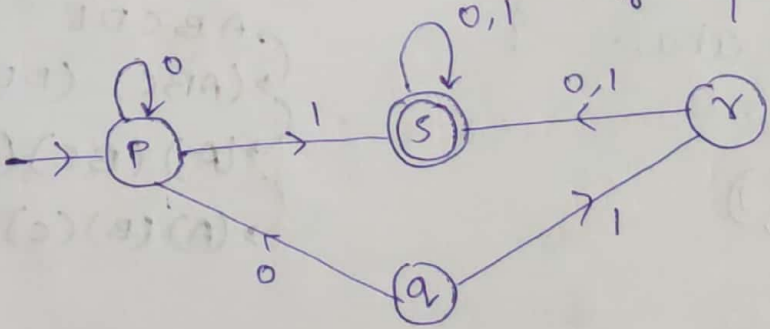
Minimise the following DFA:



STT	0	1
q ₀	q ₁	q ₃
q ₁	q ₀	q ₃
q ₂	q ₁	q ₄
q ₃	q ₅	q ₅
q ₄	q ₃	q ₃
q ₅	q ₅	q ₅

(0, 1, 2, 4) (3, 5)
 (0, 1) (2) (4) (3, 5)
 P q r s

states	0	1
P	P	S
q	P	r
S	S	S
r	S	S



* strings/where last two characters are same.

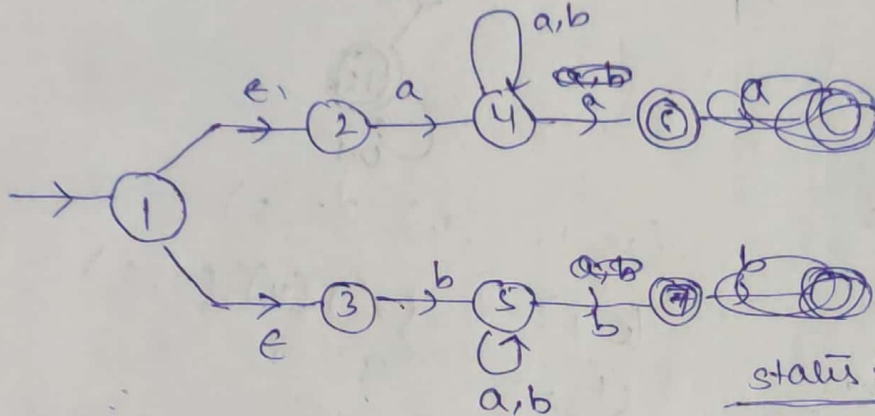
16/08/23

E-NFA conversion to DFA.

22/08/23

starting & ending with same character.

Reg exp = $a(a+b)^*a + b(a+b)^*b$



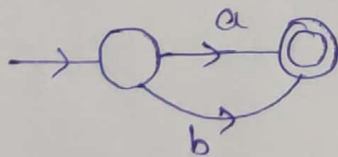
E-closure(1) = {1, 2, 3}

E-closure(4) = {4}

status ↓	a	b
→ {1, 2, 3} _A	{4} _B	{5} _C
{4} _B	{4, 6} _D	{4} _B
{5} _C	{5} _C	{5, 7} _E
* {4, 6} _D	{4, 6} _D	{4} _B
* {5, 7} _E	{5} _C	{5, 7} _E

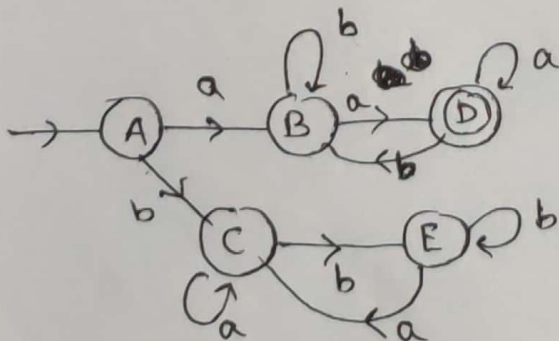
skeleton Method

$L = \{aa, bb, aba, \dots\}$



ABCDE
 (ABC) (DE)
 (A) (BC) (DE)
 (A) (B) (C) (D) (E)

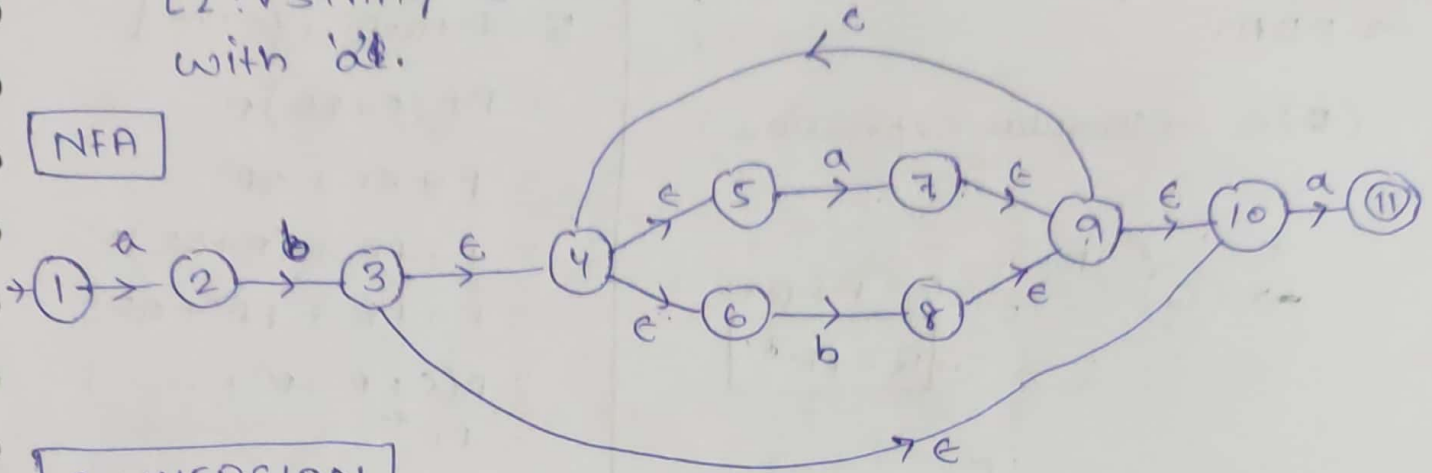
DFA



Reg Ex = $ab(a+b)^*$

L2: $\forall$ string starting with 'ab' and ending with 'a'.

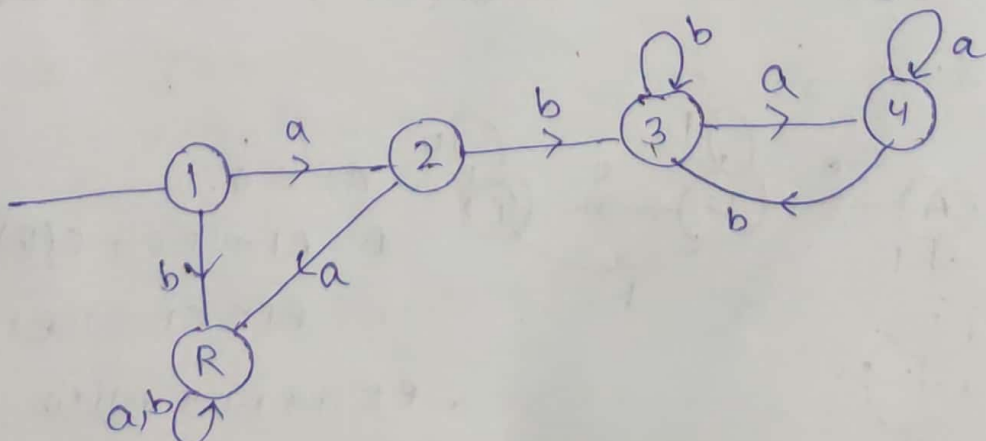
NFA



CONVERSION

State transition table

States ↓	a	b.
→ {1} A	{2} B	R F
{2} B	R F	R, 9
{3, 4, 5, 6, 8, 9, 10} C	{7, 11} D	{3, 4, 5, 6, 8, 9, 10} C
	{4, 5, 6, 7, 9, 10, 11} D	{4, 5, 6, 8, 9, 10} C
* {4, 5, 6, 7, 9, 10, 11} D	{4, 5, 6, 7, 9, 10, 11} D	{4, 5, 6, 8, 9, 10} C
6	{4, 5, 6, 7, 9, 10, 11} D	{4, 5, 6, 8, 9, 10} C
{4, 5, 6, 8, 9, 10} E	R F	R F



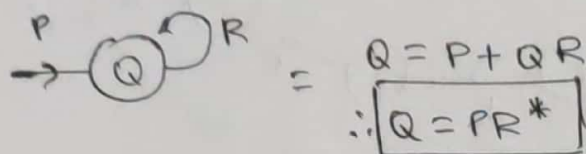
Arden's Theorem / Lemma

used to find Regular expression.

→ pumping lemma.

→ PDA.

(FA → Regular expression).



$$Q = P + QR \text{ [form]}$$

$$= P + (P + QR)R$$

$$= P + RP + QR^2$$

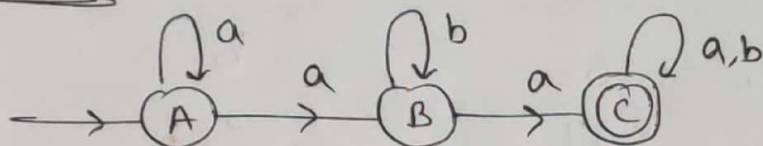
$$= P + PR + (P + QR)R^2$$

$$= P + PR + PR^2 + QR^3$$

$$\dot{=} P(\epsilon + R + R^2 + \dots)$$

$$= PR^*$$

Example 1:

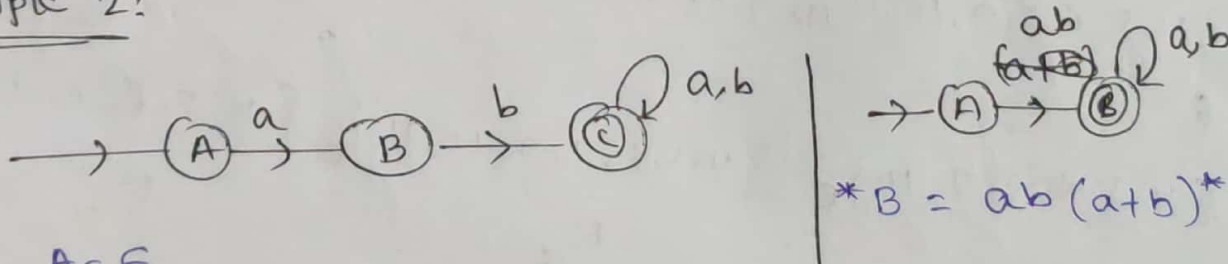


$$A = \epsilon + Aa \text{ --- (i)} \Rightarrow A = a^*$$

$$B = Aa + Bb \text{ --- (ii)} \Rightarrow B = a^*a + Bb = a^*ab^*$$

$$*C = Ba + C(a+b) \text{ --- (iii)} \Rightarrow C = (a^*ab^*)a \cdot (a+b)^*$$

Example 2:

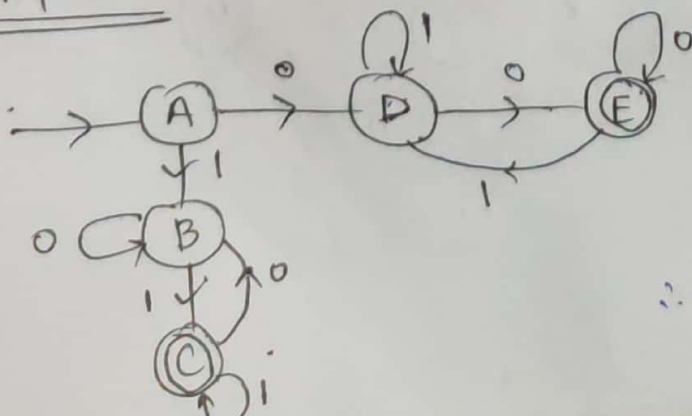


$$A = \epsilon$$

$$B = Aa = \epsilon a = a$$

$$*C = bB + C(a+b) \Rightarrow bB(a+b)^* \Rightarrow ab(a+b)^*.$$

Example 3:



$$A = \epsilon$$

$$B = A1 + B0 + C(0) \Rightarrow 1 + B0 + C0$$

$$C = B1 + C1 \Rightarrow B11^*$$

$$\therefore B = 1 + B0 + B11^*0 = 1 + B(0 + 11^*0)$$

$$= 1(0 + 11^*0)^*$$

$$\therefore C = B11^* = 1(0+11^*0)^*11^*$$

$$D = A0 + D1 + E1 \Rightarrow 0 + D1 + E1 = 0 + D1 + D00^*1$$

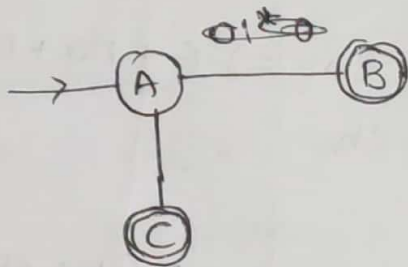
$$E = D0 + E0 = D00^*$$

$$D = 0 + D(1 + 00^*1) = 0(1 + 00^*1)^*$$

$$\therefore E = D00^* \Rightarrow E = 0(1 + 00^*1)^*00^*$$

$\therefore$ Regular expression :- $(1(0+11^*0)^*11^* + 0(1+00^*1)^*00^*)$

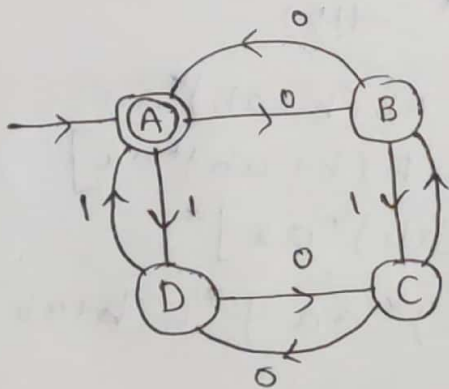
Example : 4



$$A \rightarrow B : 0(00^*1 + 1)^*00^*$$

$$B \rightarrow C : 1(11^*0 + 0)^*11^*$$

Example : 5



• No. of 0's divisible by 2.

• No. of 1's divisible by 2.

$$A = E + B0 + D1 + 1 \quad C = B1 + D0 + 1$$

$$B = A0 + C1 + 1 \quad D = A1 + C0 + 1$$

Replacing B and D in A,

$$\begin{aligned} \therefore A &= E + (A0 + C1)0 + (A1 + C0)1 \\ &= E + A00 + C10 + A11 + C01 = E + A(00 + 11) + C(01 + 10) \end{aligned}$$

Replacing B and D in C,

$$\therefore C = (A0 + C1)1 + (A1 + C0)0 = A01 + C11 + A10 + C00$$

$$C = A(01 + 10) + C(11 + 00)$$

$$\therefore C = A(01 + 10)(11 + 00)^* \quad \text{--- (vi)}$$

putting value of c in equ. (v)

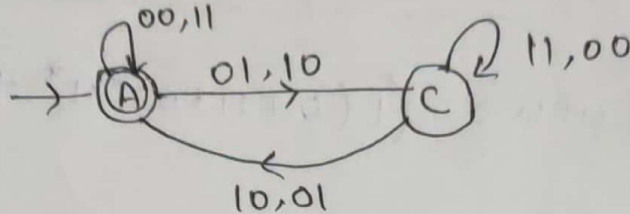
$$A = \epsilon + A(00+11) + C(01+10)$$

$$= \epsilon + A(00+11) + A(01+10)(11+00)^*(01+10)$$

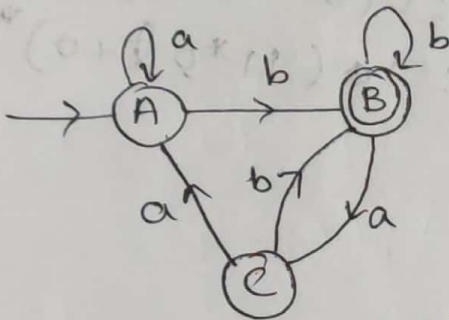
$$= \epsilon + A[(00+11) + (01+10)(11+00)^*(01+10)]^*$$

$$\therefore A = [(00+11) + (01+10)(11+00)^*(01+10)]^*$$

Example : 6



Example : 6



$$A = \epsilon + Aa + Ca \Rightarrow \epsilon + Aa + Ba a$$

$$B = Ab + Bb + Cb$$

$$C = Ba$$

$$B = Ab + Bb + Bab \Rightarrow Ab + B(b + ab)$$

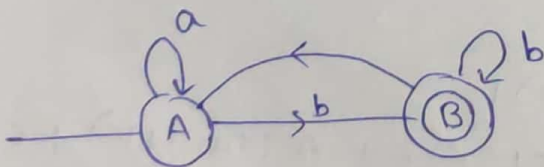
$$B = Ab(b + ab)^* \text{ (ii)}$$

$$A = \epsilon + Aa + Ab(b + ab)^* aa$$

$$A = \epsilon + A[a + b(b + ab)^* aa]$$

$$A = [a + b(b + ab)^* aa]^*$$

$$\therefore B = [a + b(b + ab)^* aa]^* b(b + ab)^*$$



$$A \rightarrow B = b(b b^*)$$

Pumping Lemma =

If any lang is Regular then:-

there will be pumping length p , and a string $w \in L$ and

we can divide $w = xyz$

such that -

i) $|xy| \leq p$

ii) $|y| > 1$

iii) for $\forall i$, $xy^iz \in L$

Example:

prove $L = \{a^n b^n \mid n \geq 1\}$ is not Regular lang.

suppose, L is Regular language.

let $p = 3$

[p is a integer value, pumping length].

$\boxed{\text{now } |w| > p}$

consider $w = a^4 b^4 = \frac{a}{x} \frac{aabb}{y} \frac{b}{z}$

y k power increase

$|xy| = 'aab' = 3 \leq 3$

✓ krna means loop banana.

$|y| > 1 \Rightarrow 'ab' = 2 > 1$ ✓

for $\forall i$, xy^iz start i with 2.

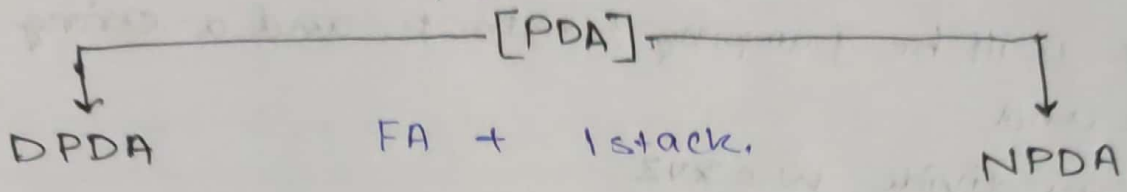
if $i = 2$ then

$xy^2z \Rightarrow a(ab)^2b = aababb \neq a^4b^4$

Hence proved this is not regular language.

How $L = \{a^p \mid p \text{ is a prime no., } \Sigma = \{a\}\}$

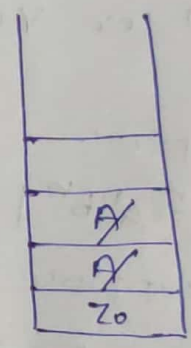
context-free language.
PUSH DOWN AUTOMATA



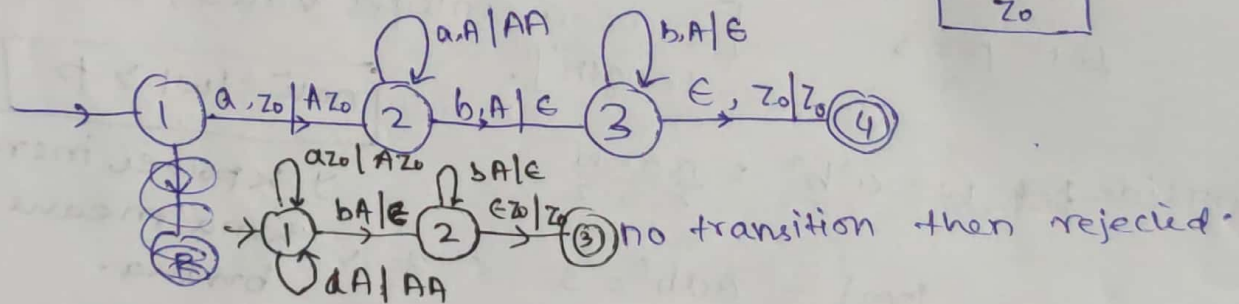
PDA = $(Q, \Sigma, \delta, q_0, F, \Upsilon, Z_0)$
 initial stack symbol.
 finite set of stack sym.
 $Z_0 \in \Upsilon$

stack - helps in counting.
 $Z_0 \rightarrow$ stack empty.

1) $L = \{a^n b^n \mid n \geq 1, \Sigma = \{a, b\}\}$
 $= \{ab, a^2b^2, a^3b^3, \dots\}$



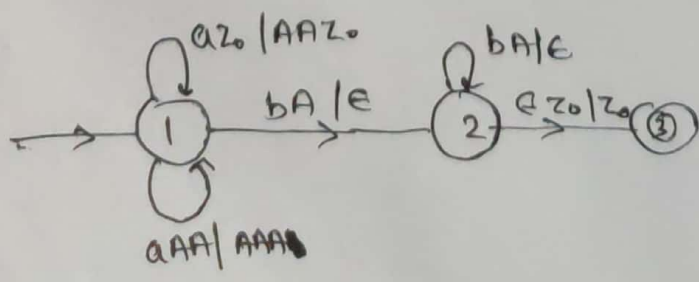
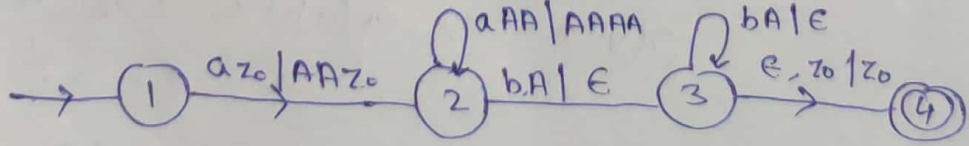
Grammar:



2) $L = \{a^n b^{2n} \mid n \geq 1, \Sigma = \{a, b\}\}$
 $= \{ab^2, a^2b^4, a^3b^6, \dots\}$

multiple insertion possible.
 pop $\rightarrow$ 1 elem. only.

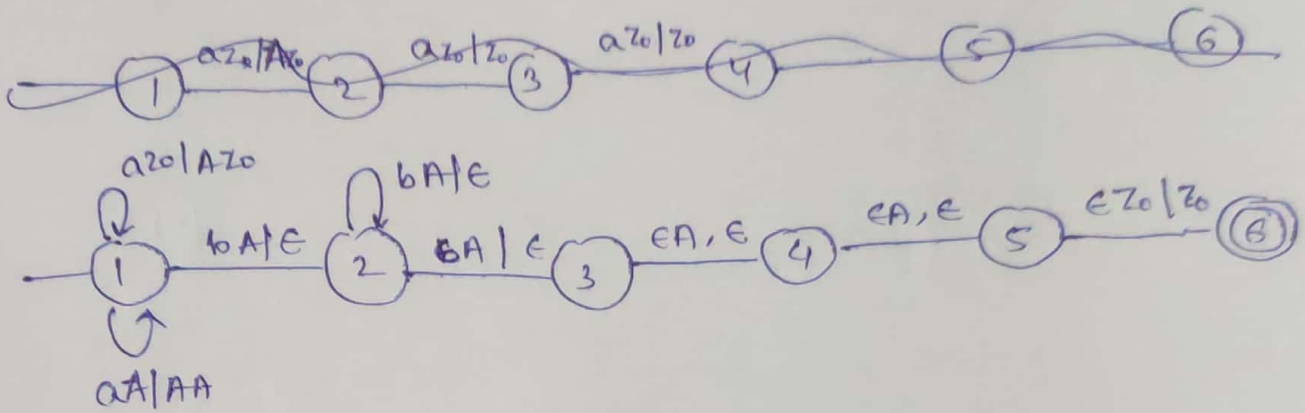
How $L = \{a^{2n} b^n \mid n \geq 1, \Sigma = \{a, b\}\}$



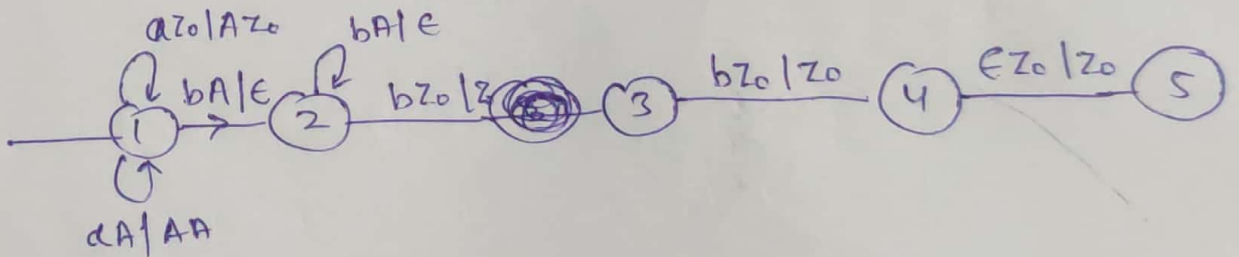
3) $L = \{ a^{n+3} b^n \mid n \geq 1, \Sigma = \{a, b\} \}$

$$\{a^4b^1, a^5b^2, a^6b^3, \dots\}$$

$$\hookrightarrow [a^3 a^n b^n]$$



4). $L = \{ a^n b^{2n} \mid n \geq 1, \Sigma = \{a, b\} \}$
 $\hookrightarrow a^n b^n b^n$



57 $L = \{a^n b^n \mid n < 10^5\}$ $\Sigma = \{a, b\}$

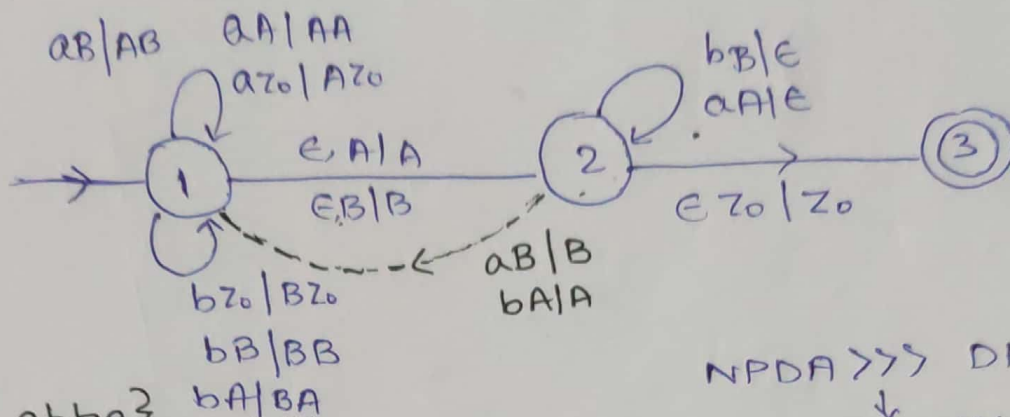
Finite or Infinite?

Finite as n is prime.

NPDA

$$= \{ \underset{1}{a} \underset{1}{b} \underset{1}{b} \text{ } \cancel{b} \cancel{a} \}$$

$$1) L = \{ ww^R \mid w \in \text{string formed by } \{a, b\} \}$$

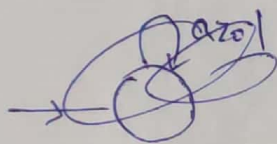


NPDA >>> DPDA
↓
powerful.

{abba abba}

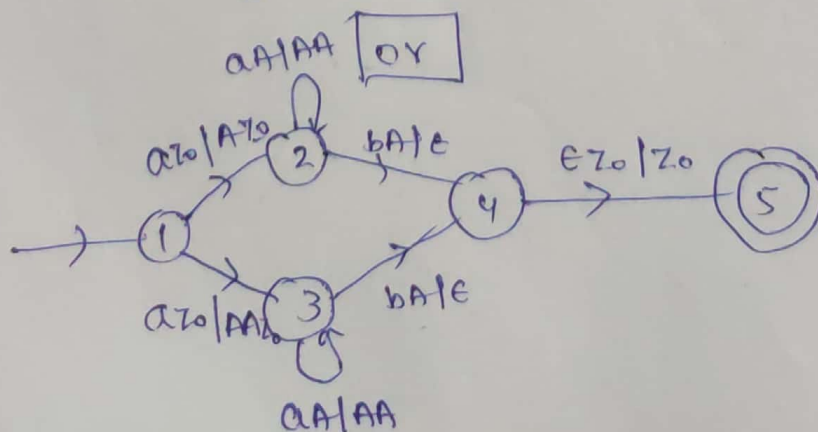
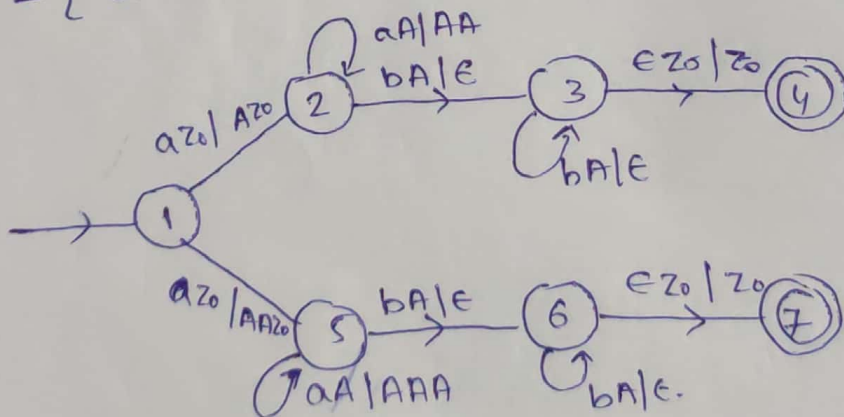
$$2) L = \{ w \text{ } \cancel{a} \text{ } cw \mid w \in \text{string formed by } \{a, b\} \}$$

= ~~{abcab}~~ Not a context-free language
crossage.



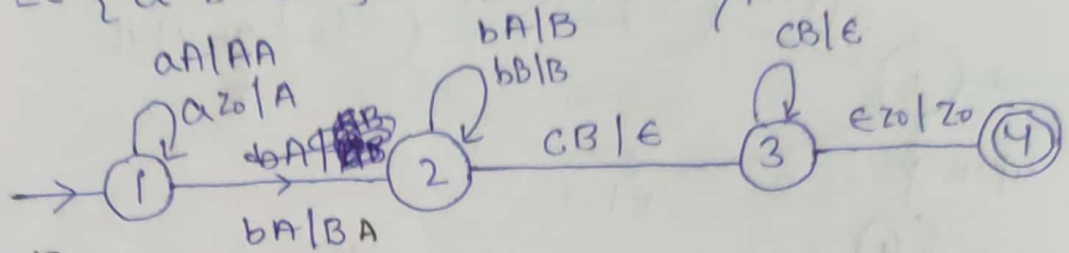
$$3) L = \{ a^n b^n \cup a^n b^{2n} \mid n \geq 1, \Sigma = \{a, b\} \}$$

$$= \{ aabb \cup aabbbb \}$$



4) $L = \{ a^n b^n c^n \mid n \geq 1, \Sigma = \{a, b, c\} \}$

$L = \{ \overbrace{a^n b^n c^n} \}$



Assignment

5) $L = \{ \text{no. of } (a) = \text{no. of } (b) \mid \Sigma = a, b \}$

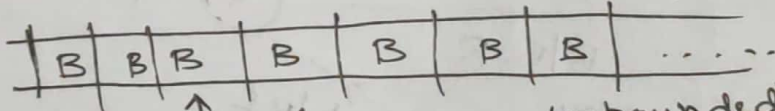
LBA

LINEAR BOUNDED AUTOMATA

Turing Machine =

TM = FA + 2 stack

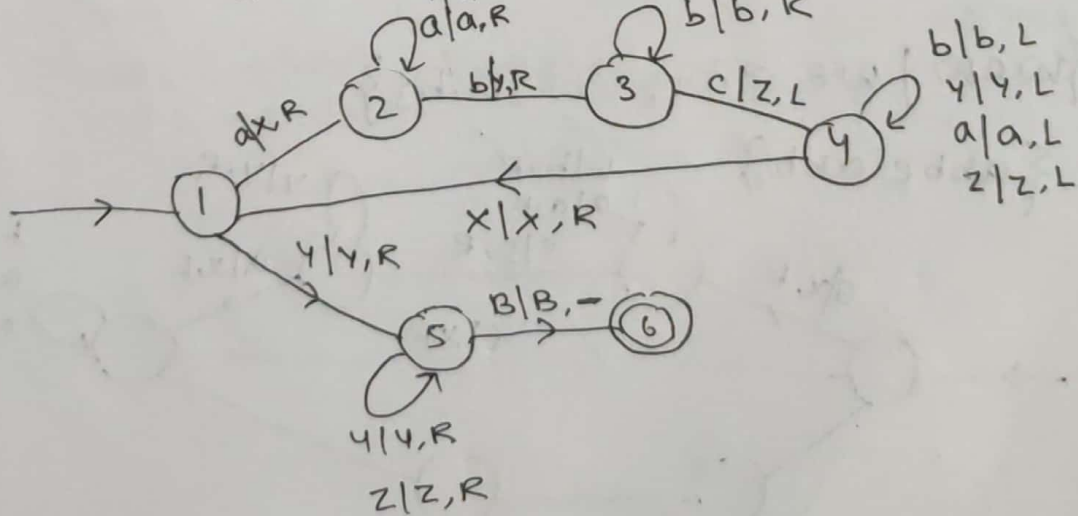
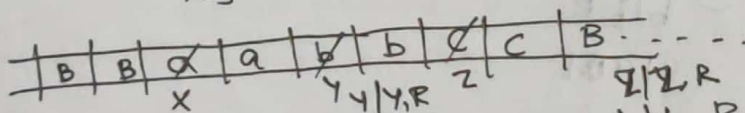
Infinite tape.



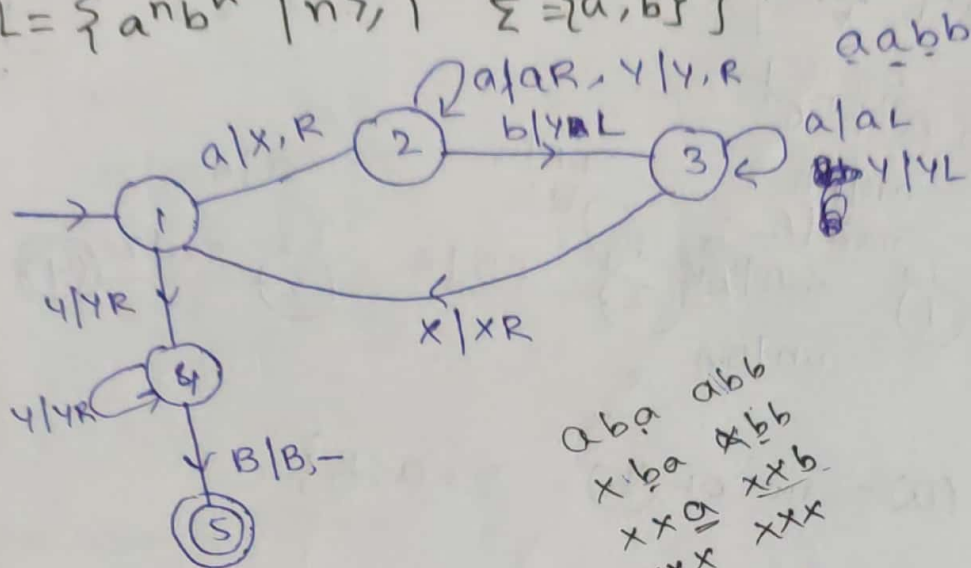
↑ head
← Left
Right →

unbounded automata.

4)



$$L = \{ a^n b^n \mid n \geq 1, \Sigma = \{a, b\} \}$$

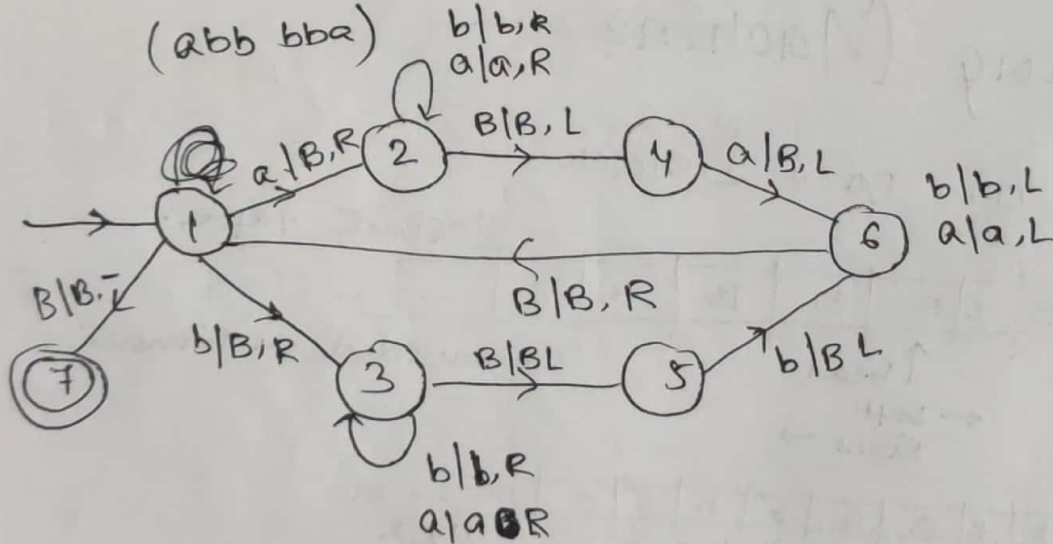


19/09/23

$$L = \{ w w^R \mid w \in \text{string using } (a, b) \}$$

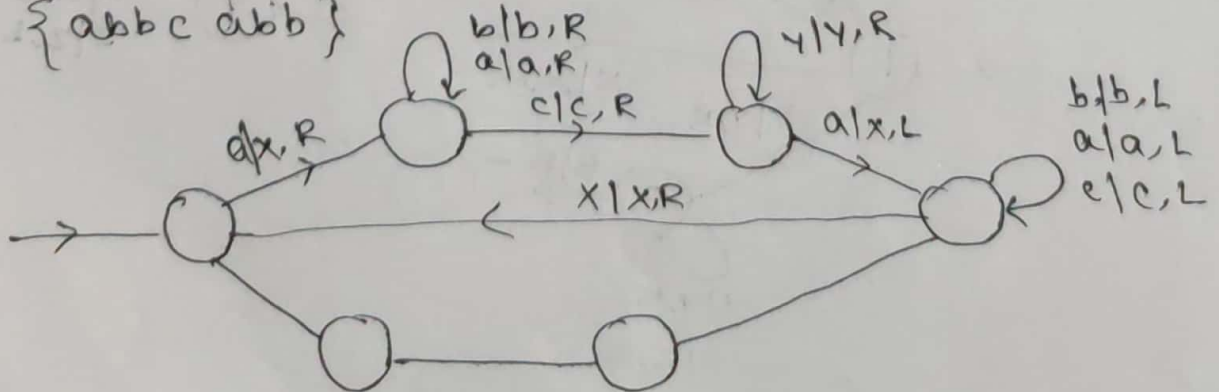
$$= \{ a^* b^* \mid \Sigma = \{a, b\} \}$$

(abb bba)

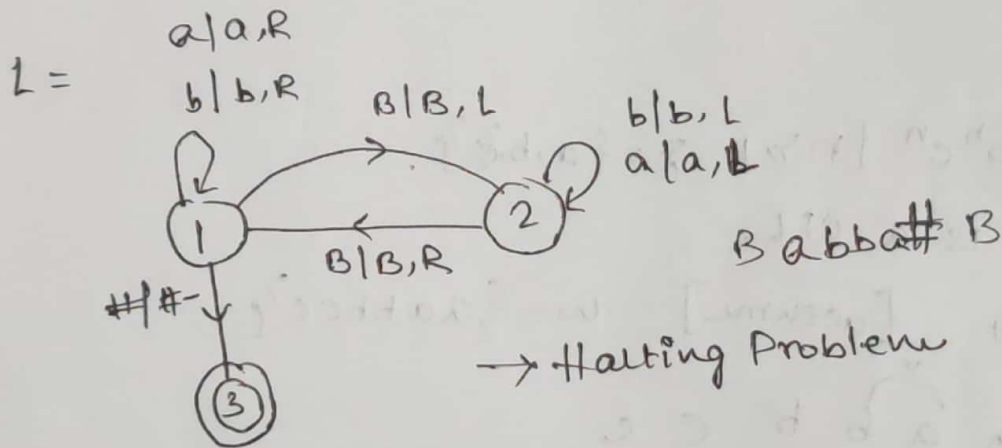
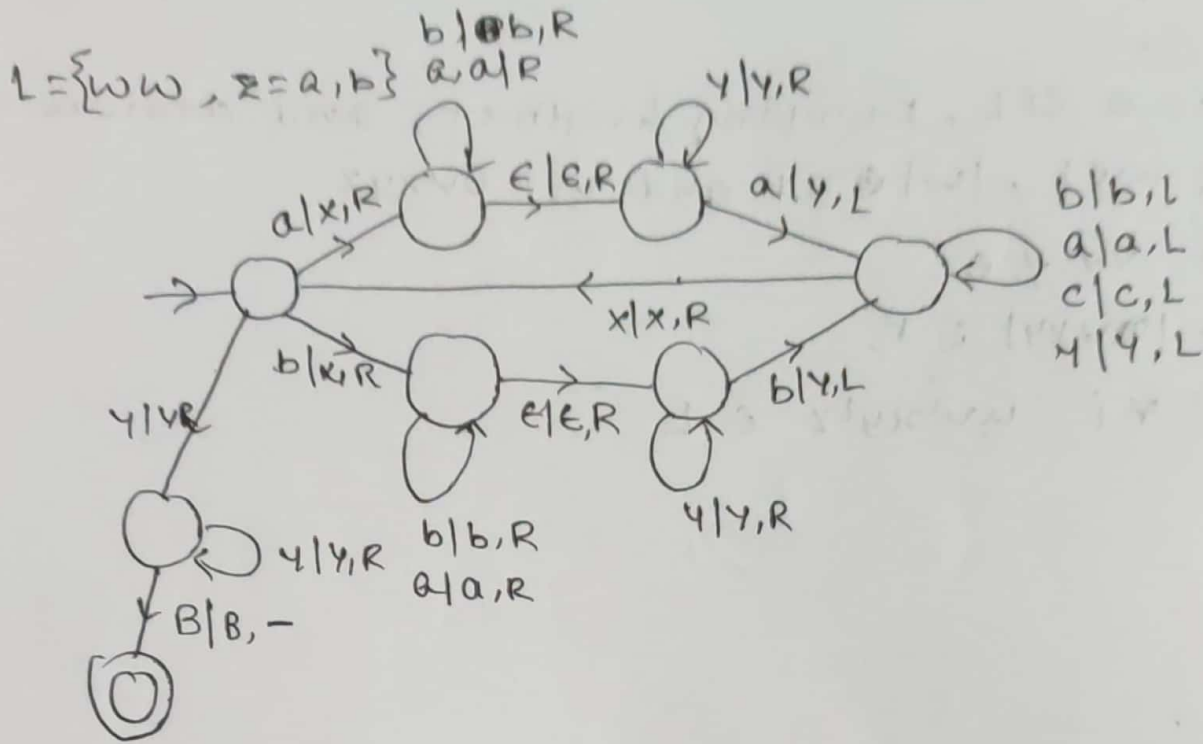


$$L = \{ w c w \mid w \in \text{string } \Sigma = \{a, b\} \}$$

{abbc abba}



* FA << DPDA <<< NPDA <<< DTM <<< NIM.



Recursive Enumerable.

TM
 $w \in L \rightarrow \text{Yes}$
 $w \notin L \rightarrow \text{No}$

} Recursive

Recursive Enumerable.

TM
 $w \in L \rightarrow \text{Yes}$
 $w \notin L \rightarrow \text{You can't say.}$

Pumping Lemma For Context Free Language :-

If L is a CFL, pumping length $= P$, and consider a string $w \in L$, $|w| \geq P$ and $w = uvxyz$

- i) $vy \neq \epsilon$,
- ii) $|vxy| \leq P$,
- iii) $\forall i, uv^i xy^i z \in L$

$$L = \{a^n b^n c^n \mid n \geq 1\}, \Sigma = \{a, b, c\}.$$

Suppose L is a CFL,

$$P=4 \quad [\text{assume}] \quad w = \{aabbcc\}$$

a a b b c c
v x y z

for $i=2$;

$$uv^2xy^2z = aababbc^2c = (a^2bab^2c^3)$$

hence not a string of $L = a^n b^n c^n$.

Grammar.

Production Rule, comb.

$N \rightarrow$ non terminal

$T \rightarrow$ Terminal.
(small letter)

at least 1 $\leftarrow T \longrightarrow (NUT)^*$
non-terminator or
connection of terminal

$ab \rightarrow NaT$ ✗

$CB \rightarrow NbT$ ✓

Regular Grammar

$N \rightarrow (NUT)^*$

constraint: only one nonterminal.

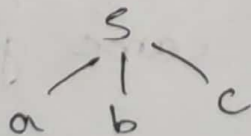
CFG

$N \rightarrow (NUT)^*$

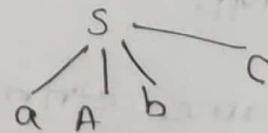
no restriction : both left and right linearity
are possible.

Q) $\begin{cases} S \rightarrow abc \\ A \rightarrow aa \end{cases}$

$\Rightarrow \begin{cases} S \rightarrow abc \\ S \rightarrow aAbc \\ A \rightarrow aa \end{cases}$



and



Ex $\rightarrow \{a^n b^n \mid n \geq 1, \Sigma = \{a, b\}\}$

CFL

$S \rightarrow asb \mid ab$

$$L = a^n b^m c^m d^n$$

31/10/23

$$\begin{cases} S \rightarrow aSd \mid aAd \\ A \rightarrow bAc \mid bc \end{cases}$$

Regular lang Grammar

$$ab(a+ b)^*$$

Q) $E \rightarrow E + E \mid E * E \mid a$ derive $a + a * a$

$$E \rightarrow E + E$$

$$\rightarrow a + E \quad [\because E \rightarrow a]$$

$$\rightarrow a + E * E \quad [\because E \rightarrow E * E]$$

$$\rightarrow a + a * E \quad [\because E \rightarrow a]$$

$$\rightarrow a + a * a \quad [\because E \rightarrow a]$$

→ LMD - left most derivation
Evaluate left most NT.

Ambiguous Grammar: If there's possibility to solve a string with more than

1) LMD or 3) LMD and RMD.

2) RMD

* If same string generates same grammar and follows this constraint.

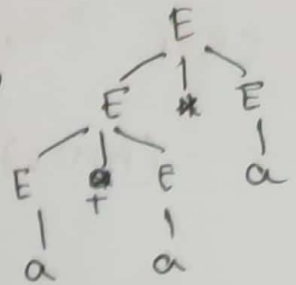
If there exist more than one distinct ~~grammars~~ derivation tree then it's ambiguous.

* Now we have to remove it.

$$E \rightarrow E * E$$

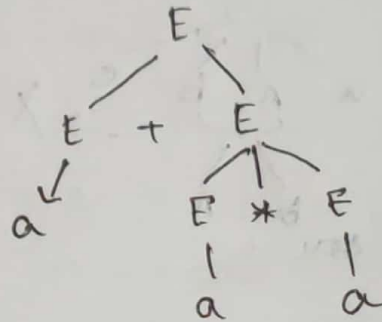
$$E \rightarrow E + E \quad [E \rightarrow E + E]$$

$$E \rightarrow E + E * a$$

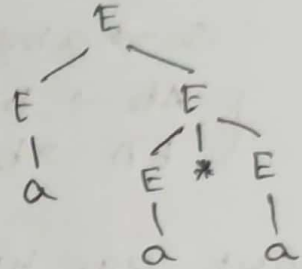


Derivation tree.

LMD



RMD



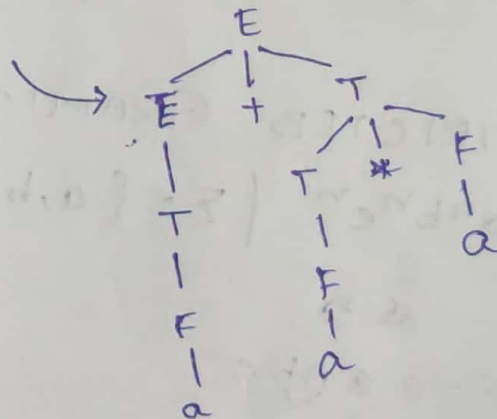
Derivation tree or parse tree.

Removing Ambiguity :-

$$E \rightarrow E + E \mid E * E \mid a$$

Always start with start symbol i.e. E

$$\left\{ \begin{array}{l} E \rightarrow E + T \mid T \\ T \rightarrow T * F \mid F \\ F \rightarrow a \end{array} \right.$$



SIMPLIFICATION OF CFG

Left Recursion

$$\begin{cases} E \rightarrow E + T \mid T \\ T \rightarrow T * F \mid F \\ F \rightarrow a \end{cases}$$

if LHS = RHS first NT elem
then it's left recursion

Formula

with left recy.

$$A \rightarrow A \alpha \mid B$$

without left recursion.

α and B production rule.

Soln:

$$A \rightarrow B A'$$

$$A' \rightarrow \alpha A' \mid \epsilon$$

$$A \rightarrow A \alpha_1 \mid A \alpha_2 \mid B_1 \mid B_2 \mid B_3$$

$$A \rightarrow B_1 A' \mid B_2 A' \mid B_3 A'$$

$$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \epsilon$$

$S \rightarrow ies | ieses$ [from left side]..

factoring.

$S \rightarrow ies S'$

$S' \rightarrow es / \epsilon$

(A) NULL PRODUCTION REMOVAL:-

$S \rightarrow ABaD$

$A \rightarrow BD$

$B \rightarrow a | \epsilon$

$D \rightarrow b | \epsilon$

$\rightarrow$ LHS mein null hai
Nullable : not terminals. A, B, D

$A \rightarrow BD | D | B$

$B \rightarrow a$

$D \rightarrow b$

Example :-

$S \rightarrow ABaD$

$A \rightarrow BD$

$B \rightarrow a | \epsilon$

$D \rightarrow b$

Nullable : non-terminal.

$S \rightarrow ABaD | AaD$

$A \rightarrow BD | D$

$B \rightarrow a$

$D \rightarrow b$

(5) REMOVAL OF UNIT PRODUCTION

$A \rightarrow D$

$A \rightarrow D$

$B \rightarrow b$

LHS mein there is only one non-terminal

To remove unit production:-

$S \rightarrow aAb$

$A \rightarrow B$

$B \rightarrow d$

$S \rightarrow aAb$

$A \rightarrow d$

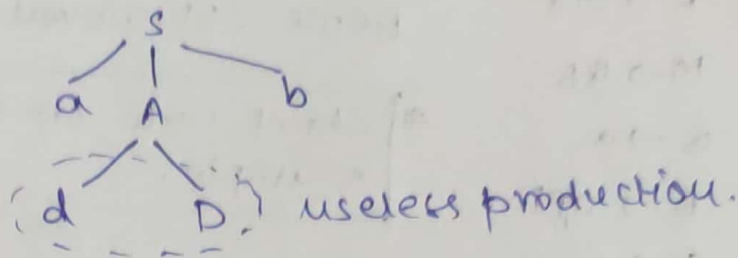
$B \rightarrow d$

useless production rule.

(6) REMOVAL OF USELESS PRODUCTION :

$$S \rightarrow aAb$$

$$A \rightarrow dD/b$$



Q) Simplify this Grammar.

$$S \rightarrow aA | aBB$$

$$A \rightarrow aaA | \epsilon$$

$$B \rightarrow bB | bbD$$

$$D \rightarrow B$$

$$S \rightarrow aA | a | aBB$$

$$A \rightarrow aaA | aa$$

$$B \rightarrow bB | bbD$$

$$\boxed{D \rightarrow B} \text{ — } D \rightarrow bB | bbD.$$

useless production

$$S \rightarrow aBB$$

$$B \rightarrow bB | bbD$$

$$D \rightarrow bB | bbD$$

$\Rightarrow$

$$S \rightarrow aA | a$$

$$A \rightarrow aaA | aa$$

Find Answer

NORMAL FORMS :-

eg. $A \rightarrow aBAD$

$$B \rightarrow a | PD$$

$$D \rightarrow a$$

$$P \rightarrow b | \epsilon$$

to convert CNF or GNF

1) Null Production

2) Unit Production

3) Useless Production

eg. $A \rightarrow aBAD$

$$B \rightarrow a | PD$$

$$D \rightarrow a$$

$$D \rightarrow b$$

$\Rightarrow$

$$A \rightarrow aBAD$$

$$Q \rightarrow a$$

$$B \rightarrow a | PD$$

$$D \rightarrow a$$

$$P \rightarrow b$$

$\Rightarrow$

$$A \rightarrow MQD$$

$$M \rightarrow QB$$

⇒ $A \rightarrow RD$

$R \rightarrow MR$

$M \rightarrow RB$

$B \rightarrow a$

$R \rightarrow dR$

$D \rightarrow a$

$F \rightarrow b$

Hence CFG converted.

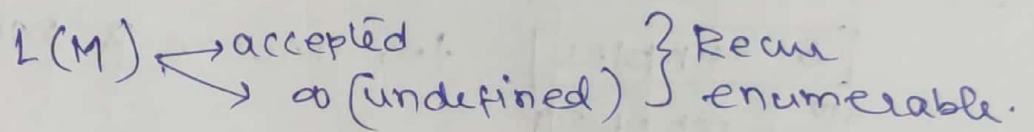
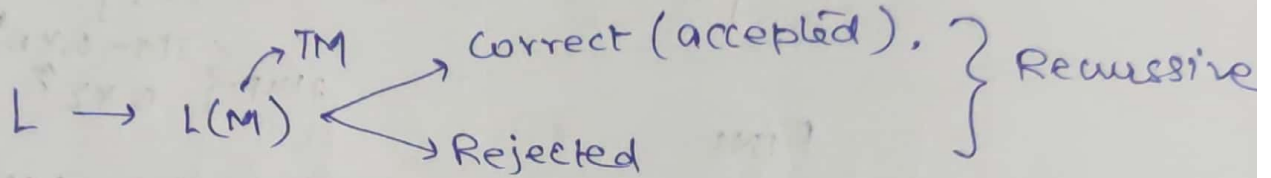
* Short note on CFG and GNF with example.

IMPORTANT

lang formation.

Formal lang.

Recursive enumerable.
enumerable



Grammar + derivations (LMD, RMD)

